PROBLEM 20

# Heavy Particle on a Rotating Circumference

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Energy in a Rotating Frame

A heavy particle is moving along a smooth circumference of radius $R$ and center $O$. One of the diameters of the circumference is along the local vertical, and the circumference is rotating with constant angular velocity $\omega$ about it. The position of the particle with respect to the circumference is described by the angle $\theta$ between the particle and the lowest point of the circumference.

**(a)** Discuss the conservation of mechanical energy in the moving reference frame and compute its value.

**(b)** Compute the reaction of the circumference in the direction normal to its plane, as a function of $\theta$.

### Worked Solution (4-Phase Methodology)

PHASE 1

#### Physical Statement, Hypotheses & Parameters

- **Physical system:** a particle $P$ of mass $m$ slides on a smooth circular wire of radius $R$ and centre $O$; the wire turns with constant angular velocity $\omega$ about its vertical diameter. The angle $\theta$ is measured from the lowest point.
- **Why the moving frame.** In the frame $S_1$ attached to the wire the constraint is a fixed circle, the motion has one coordinate $\theta$, and the reaction has two clear components (in the plane of the wire, normal to it). The question itself asks for the energy "in the moving

reference frame", which requires the inertia forces of Notes Eq. 4.62 and their potentials (Slides 29–31).

- **Hypotheses:**
  1. $S_0$ fixed to the ground is inertial; $O$ is fixed; uniform gravity $-g\,\mathbf{k}$.
  2. The wire is smooth: the reaction has no component along the tangent of the wire.
  3. $\omega$ is constant.
- **Degrees of freedom:** the particle is constrained to a curve fixed in $S_1$: **1 DOF**, $\theta$.
- **Unknown reaction:** $\mathbf{N} = N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1$, with $N_r$ in the plane of the wire (along the radius) and $N_b$ normal to that plane.

| Symbol | Meaning | SI unit |
|---|---|---|
| $m$ | mass of the particle | kg |
| $R$ | radius of the circumference | m |
| $\omega$ | spin rate of the circumference (constant) | rad/s |
| $g$ | gravitational acceleration | m/s$^2$ |
| $\theta$ | angle from the lowest point | rad |
| $N_r,\ N_b$ | reaction components (radial in-plane, normal to the plane) | N |
| $\theta_0,\ \dot\theta_0$ | initial angle and angular rate in $S_1$ | rad, rad/s |

PHASE 2

### Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, $\mathbf{k}_0$ upward along the rotation axis. $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire: $\mathbf{k}_1 = \mathbf{k}_0$, $\mathbf{i}_1$ horizontal in the plane of the wire (on the side where $\theta > 0$), $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ normal to the plane of the wire. With the wire plane at azimuth $\omega t$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Motion of $S_1$:** $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$.
- **Wire basis.** $\mathbf{e}_r = \sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{k}_1$ (from $O$ to $P$), $\mathbf{e}_\theta = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1$ (tangent, increasing $\theta$). Since $\mathbf{e}_r \times \mathbf{e}_\theta = (-\cos\theta)(\cos\theta)(\mathbf{k}_1 \times \mathbf{i}_1) + (\sin\theta)(\sin\theta)(\mathbf{i}_1 \times \mathbf{k}_1) = -\mathbf{j}_1$, the right-handed triad is $\{\mathbf{e}_r, \mathbf{e}_\theta, -\mathbf{j}_1\}$:

$$[{}_1R_w] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \\ -\cos\theta & \sin\theta & 0 \end{pmatrix}, \qquad \det[{}_1R_w] = \sin\theta(0\cdot 0 + \sin\theta) - \cos\theta(0\cdot 0 - \cos\theta) = 1$$

and the columns are orthonormal ($\sin^2\theta + \cos^2\theta = 1$, $\sin\theta\cos\theta - \cos\theta\sin\theta = 0$), so $[{}_1R_w][{}_1R_w]^T = I$.

- **Derivatives in $S_1$** (chain rule): $\frac{d\mathbf{e}_r}{dt}\big|_1 = \dot\theta\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1) = \dot\theta\,\mathbf{e}_\theta$ and $\frac{d\mathbf{e}_\theta}{dt}\big|_1 = \dot\theta\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{k}_1) = -\dot\theta\,\mathbf{e}_r$.

PHASE 3

## Step-by-Step Mathematical Deduction

**Relative kinematics and forces in $S_1$**

$$\mathbf{r}_1^P = R\,\mathbf{e}_r, \qquad \mathbf{v}_1^P = R\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_1^P = R\ddot\theta\,\mathbf{e}_\theta + R\dot\theta\left(-\dot\theta\,\mathbf{e}_r\right) = R\ddot\theta\,\mathbf{e}_\theta - R\dot\theta^2\,\mathbf{e}_r$$

(product rule on $R\dot\theta\,\mathbf{e}_\theta$). Forces seen from $S_1$ (Notes Eqs. 4.62–4.63):

- weight $-mg\,\mathbf{k}_1$ and reaction $\mathbf{N} = N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1$ (actual forces);
- $\mathbf{F}_A = \mathbf{0}$ ($O$ fixed) and $\mathbf{F}_E = \mathbf{0}$ ($\boldsymbol{\alpha}_{10} = \mathbf{0}$);
- centrifugal force: $\mathbf{r}_\perp = R\sin\theta\,\mathbf{i}_1$ (the part of $\mathbf{r}_1^P$ normal to $\mathbf{k}_1$), so (Slide 30)

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = m\omega^2 R\sin\theta\,\mathbf{i}_1$$

Component check: $\omega\mathbf{k}_1 \times (R\sin\theta\,\mathbf{i}_1 - R\cos\theta\,\mathbf{k}_1) = R\omega\sin\theta\,\mathbf{j}_1$, then $\omega\mathbf{k}_1 \times R\omega\sin\theta\,\mathbf{j}_1 = -R\omega^2\sin\theta\,\mathbf{i}_1$, and the minus sign of $\mathbf{F}_C$ gives $+m\omega^2 R\sin\theta\,\mathbf{i}_1$;

- Coriolis force, with $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\mathbf{F}_{COR} = -2m\,\omega\mathbf{k}_1 \times R\dot\theta\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1) = -2m\omega R\dot\theta\cos\theta\,\mathbf{j}_1$$

**Equation of motion** (projection on $\mathbf{e}_\theta$, where $\mathbf{N}$ has no component). Using $\mathbf{k}_1 \cdot \mathbf{e}_\theta = \sin\theta$, $\mathbf{i}_1 \cdot \mathbf{e}_\theta = \cos\theta, \mathbf{j}_1 \cdot \mathbf{e}_\theta = 0$:

$$mR\ddot{\theta} = -mg\sin\theta + m\omega^2 R\sin\theta\cos\theta \quad \Longrightarrow \quad \ddot{\theta} = -\frac{g}{R}\sin\theta + \omega^2\sin\theta\cos\theta$$

**(a) Mechanical energy in the moving frame**

- **Justification:** $E_1^P$ is conserved when every force that does work in $S_1$ derives from a potential (Slide 29). We check the forces one by one using their power in $S_1$, $\mathbf{F}\cdot\mathbf{v}_1^P$.
- **Reaction:** $\mathbf{N}\cdot\mathbf{v}_1^P = R\dot{\theta}\,(N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1)\cdot\mathbf{e}_\theta = 0$: the wire is smooth and **fixed in** $S_1$, so it does no work in $S_1$.
- **Coriolis force:** $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ always (Slide 31); here explicitly $\mathbf{j}_1\cdot\mathbf{e}_\theta = 0$.
- **Euler force and origin force:** both zero.
- **Weight:** conservative, $V_g = mgz_1 = -mgR\cos\theta$ (zero at the height of $O$).
- **Centrifugal force:** $\boldsymbol{\omega}_{10}$ is constant, so it has the potential (Slide 30) $V_1 = -\frac{1}{2}m\omega^2 r_\perp^2 = -\frac{1}{2}m\omega^2 R^2\sin^2\theta$ (zero on the axis).

Therefore $E_1^P$ **is conserved**:

$$\boxed{E_1^P = \frac{1}{2}mR^2\dot{\theta}^2 - mgR\cos\theta - \frac{1}{2}m\omega^2R^2\sin^2\theta = \frac{1}{2}mR^2\dot{\theta}_0^2 - mgR\cos\theta_0 - \frac{1}{2}m\omega^2R^2\sin^2\theta_0}$$

its value being fixed by the initial conditions $\theta(0) = \theta_0, \dot{\theta}(0) = \dot{\theta}_0$ (the statement gives none, so the constant is left in terms of them).

**Check by differentiation** (chain rule on each term, then the EOM):

$$\frac{dE_1^P}{dt} = mR^2\dot{\theta}\ddot{\theta} + mgR\sin\theta\,\dot{\theta} - m\omega^2R^2\sin\theta\cos\theta\,\dot{\theta} = mR^2\dot{\theta}\left[\ddot{\theta} + \frac{g}{R}\sin\theta - \omega^2\sin\theta\cos\theta\right] = 0$$

**The energy in $S_0$ is not conserved.** In $S_0$ the wire moves, and its reaction does work: with $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = R\dot{\theta}\,\mathbf{e}_\theta + \omega R\sin\theta\,\mathbf{j}_1$ (Notes Eq. 3.26), the power of $\mathbf{N}$ in $S_0$ is $\mathbf{N}\cdot\mathbf{v}_0^P = N_b\,\omega R\sin\theta \neq 0$. This is why the moving frame is the natural one for an energy integral here.

**(b) Reaction normal to the plane of the circumference**

> **Pedagogical Justification:** project Notes Eq. 4.63 on $\mathbf{j}_1$, normal to the plane. The relative motion stays in the plane, so $\mathbf{a}_1^P \cdot \mathbf{j}_1 = 0$; the weight and the centrifugal force lie in the plane; only $N_b$ and the Coriolis force remain.

$$m\,\mathbf{a}_1^P \cdot \mathbf{j}_1 = 0 = N_b - 2m\omega R\dot{\theta}\cos\theta \quad \Longrightarrow \quad N_b = 2m\omega R\dot{\theta}\cos\theta$$

To express it as a function of $\theta$, solve the energy integral of part (a) for $R\dot{\theta}$:

$$R^2\dot{\theta}^2 = \frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2 R^2\sin^2\theta \quad \Longrightarrow \quad R\dot{\theta} = \pm\sqrt{\frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2 R^2\sin^2\theta}$$

$$\boxed{N_b(\theta) = \pm\,2m\omega\cos\theta\,\sqrt{\frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2 R^2\sin^2\theta}}$$

with the sign of $\dot{\theta}$ (positive when $\theta$ increases) and $E_1^P$ from part (a).

**In-plane component (for completeness).** Projecting on $\mathbf{e}_r$ ($\mathbf{k}_1 \cdot \mathbf{e}_r = -\cos\theta$, $\mathbf{i}_1 \cdot \mathbf{e}_r = \sin\theta$):

$$-mR\dot{\theta}^2 = N_r + mg\cos\theta + m\omega^2 R\sin^2\theta \quad \Longrightarrow \quad N_r = -mR\dot{\theta}^2 - mg\cos\theta - m\omega^2 R\sin^2\theta$$

PHASE 4

### Physical Interpretation & Dimensional Verification

- **Origin of $N_b$.** The normal-to-plane reaction is entirely due to the Coriolis force: the wire must push the particle sideways so that it keeps turning with the plane while its distance to the axis changes. It vanishes when $\dot{\theta} = 0$ and when $\cos\theta = 0$ (the particle crosses the horizontal diameter, where $\mathbf{v}_1^P \parallel \boldsymbol{\omega}_{10}$).

- **Relative equilibria** (extension, from $\ddot{\theta} = 0$ with $\dot{\theta} = 0$): $\sin\theta\left(\omega^2\cos\theta - g/R\right) = 0$, i.e. $\theta = 0, \theta = \pi$, and, if $\omega^2 R > g$, $\cos\theta = g/(\omega^2 R)$; at all of them $N_b = 0$.
- **Limit $\omega \to 0$:** $N_b \to 0$ and $E_1^P \to \frac{1}{2}mR^2\dot{\theta}^2 - mgR\cos\theta$, the energy of the simple pendulum.
- **Dimensions:** each term of $E_1^P$ is $\mathrm{kg\,m^2\,s^{-2}} = \mathrm{J}$; $[N_b] = [m\omega R\dot{\theta}] = \mathrm{kg\,s^{-1}\,m\,s^{-1}} = \mathrm{N}$; inside the root, $[2E_1^P/m] = [gR] = [\omega^2 R^2] = \mathrm{m^2\,s^{-2}}$.
- **Verification (sympy, `verify_P20.py`):** the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition; solving $m\mathbf{a}_0^P = -mg\mathbf{k} + \mathbf{N}$ in $S_0$ returns the same EOM, $N_b$ and $N_r$; $dE_1^P/dt = 0$ along the EOM, while $dE_0^P/dt$ equals the power $N_b\,\omega R\sin\theta$ of the reaction in $S_0$.
- **Cross-check with Topic 3.** The Topic 3 solution gives the same $E'$ (there called the effective energy) and $N_\phi = 2m\omega R\dot{\theta}\cos\theta$. No official key exists for this problem.

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